

Sudip Niroula

Full-Stack Developer

Birtamod, Nepal | open to remote

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Summary

Full-stack developer who ships products end to end, from the data model up to the screen users touch. Five products live in production this past year, including a multi-tenant healthcare platform, a public booking system across four locations, and a mobile app on both app stores. Next.js and TypeScript on the front, Python and FastAPI behind it. Comfortable putting LLMs inside the product, not just alongside it.

Experience

Full-Stack Developer, Nullion Technologies

Apr 2026 – Present

Nepal, remote

- **AlignCare Hub (live)**: multi-tenant operations console for U.S. home-care agencies. Built tenant isolation, auth, the data model, and the reporting layer agency staff use every day.
- **Ohio WebCheck Scheduler (live)**: public booking system for FBI and BCI fingerprint background checks. Live availability across four locations, no account required, confirmation-code lookup for reschedules. Most bookings finish in under two minutes.
- **Cafe 8848 (live)**: ordering and tiered loyalty platform, shipped as native **iOS** and **Android** apps alongside web.
- **Zoe's Beer (live)** and **Nullion Bookkeeping**: a marketplace with inventory management, and a bookkeeping platform with ledgers, trial balance, profit and loss, and balance sheet.
- Stack throughout: Next.js and TypeScript on the front, FastAPI and PostgreSQL behind it, Docker for deploys.

Software Engineer, FutureFrame Pvt. Ltd.

Feb 2025 – Mar 2026

Lalitpur, Nepal

- Built **Football Score Pro**, a match management system: PySide6 desktop app with a React and FastAPI control room, structured with dependency injection and an event bus so services stayed decoupled.
- Built **Football Score Lite**, live broadcast graphics for streamed matches. PIXI.js renderers drop into OBS as browser sources, kept in sync with the operator over WebSockets.
- Maintained real-time pipelines syncing a Python backend to a WebGL frontend at sub-second latency, and replaced a manual data-ingestion process with multi-agent AI workflows.

Building with LLMs

- **Fingerprint (code)**: real-time multimodal AI tutor for children, built for the Google Gemini API Developer Competition. Gemini Live handles voice while a backend planner decides in real time whether to show a visual, run a quiz, search, or teach from a worksheet the learner uploads.
- **In-browser assistant** on sudipniroula.com.np: runs Qwen2.5 0.5B fully client-side via WebGPU, no server and no API key. Keyword retrieval feeds only the relevant facts into a short context, which is what makes a sub-1B model actually usable.
- **Hybrid retrieval system (code)**: BM25 with a hand-built inverted index plus TF-IDF and SVD, combined through reciprocal rank fusion. Evaluated with MAP, MRR and nDCG. Found and fixed a query-alignment bug in the public dataset that would otherwise have invalidated every result.

Selected Independent Work

- **Rustle (code)**: Windows search launcher written in Rust. Fuzzy matching handles typos, Rayon does parallel indexing with a roughly 150ms startup scan.
- **Labhya Compute (code)**: GPU rental marketplace with containerized SSH access to remote machines and a relay server routing requests between providers and renters. *Winner, Most Innovative Project, Ambition Hackathon 2025.*

Technical Skills

Languages: TypeScript, JavaScript, Python, Rust, SQL

Frontend: Next.js, React, Tailwind CSS

Backend: FastAPI, Node.js, Django REST, PostgreSQL, WebSockets

AI: Gemini API, LangChain, WebLLM, retrieval and evaluation, Cursor and Claude Code

Infrastructure: Docker, AWS, GCP, CI/CD, Git

Education

BSc, Computer Science and Information Technology

2022 – 2026

Mechi Multiple Campus, Tribhuvan University, Nepal